

bindsight: a reproducible bridge from RNA-seq counts to *de novo* protein binder design

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Abstract

Computational biology has two parallel ecosystems that rarely talk to each other. **RNA-seq differential-expression analysis** reliably identifies genes dysregulated in disease but stops at the gene list. **De novo protein-binder design** produces binders against arbitrary protein targets but assumes the user has already chosen the target. Bridging the two — “*this gene is up in disease, low in healthy tissue, surface-exposed, and here is a designed binder candidate ranked by predicted affinity*” — is currently an ad-hoc, project-specific exercise. We present **bindsight**, the first open-source command-line tool and web application that ingests RNA-seq counts and outputs ranked, structurally- validated *de novo* protein-binder candidates with a complete PROV-O / RO-Crate audit trail back to the patient cohort the targets came from. The discovery half of the pipeline (differential expression → surfaceome filter → druggability → AlphaFoldDB structure pull) runs on a CPU laptop; the GPU half (RFDiffusion backbone → ProteinMPNN sequence → Boltz-2 validation) executes end-to-end on free Google Colab notebooks, paid Modal serverless GPU, local NVIDIA Docker, or Kaggle. A public web demo at bindsight.streamlit.app runs the full discovery pipeline in any browser without installation. On a real TCGA breast-carcinoma cohort (tumor vs. adjacent normal, auto-downloaded from NIH/GDC), the pipeline discovers antibody-tractable cell-surface antigens over-expressed in tumor with full provenance; established targets such as ERBB2 (HER2) surface among the candidates when their expression signal is present. Software is MIT-licensed and archived at Zenodo with permanent DOI [10.5281/zenodo.20121496](https://doi.org/10.5281/zenodo.20121496).

1 Introduction

The pace of progress in *de novo* protein-binder design has been extraordinary. RFDiffusion [1] and ProteinMPNN [2] demonstrated that diffusion-based backbone generation paired with deep-learning sequence design can produce experimentally validated binders against user-specified targets at a fraction of historical cost. BindCraft [3] extended these ideas with one-shot pipelines that report 10–100% experimental success rates. Boltz-2 [4] and BoltzGen [5] added fast, accurate, and now permissively licensed structure-and-affinity prediction. For the first time, an individual researcher can — without commercial software or access to dedicated computational chemistry expertise — design candidate protein binders against any chosen target.

Choosing the target, however, remains where the field is least automated. Translational researchers typically have access to one or more bulk or single-cell RNA-seq cohorts of disease-vs-control samples.

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Standard analyses with DESeq2 [6] or edgeR [7] produce a ranked list of differentially expressed genes. Picking from that list a target that is (a) surface-exposed, (b) specific to the disease tissue, (c) druggable with the modality of interest (*e.g.*, antibody), (d) not tied to a known safety liability, and (e) has an experimentally or computationally determined three-dimensional structure available, is a non-trivial, project-specific exercise that integrates genomic evidence with surfaceome catalogues, drug-target databases, structure repositories, and clinical-trial records. In practice this work is done ad hoc, takes a competent graduate student several weeks, and is rarely reproducible across labs.

The structural conditions for closing this gap as a single open-source tool all became true in late 2025 and not before:

1. The **SURFACE-Bind catalogue** [8] published pre-computed targetable interfaces and binder seeds for $\sim 2,800$ human cell-surface proteins, removing the per-target epitope-prediction bottleneck.
2. **Permissive licensing** for state-of-the-art structure-and- affinity predictors became the norm; Boltz-2 and BoltzGen released both code and weights under MIT, removing the previously universal non-commercial restriction inherited from AlphaFold2 weights.
3. **Free GPU tiers** (Google Colab T4, Kaggle T4 $\times 2$) became powerful enough to run RFdiffusion and ProteinMPNN at meaningful scale, removing the institutional-cluster barrier.

We present **bindsight** (v0.1.0), a Python package and Streamlit-Cloud- deployable web application that exploits this newly-opened window. It takes RNA-seq counts as input and produces ranked *de novo* protein- binder candidates as output, with a complete W3C PROV-O [9] JSON-LD provenance graph and an RO-Crate [10] packaged artifact for direct deposit to Zenodo.

2 Methods and architecture

2.1 Pipeline overview

bindsight's discovery half is a Snakemake [11]-orchestrated, Pydantic-typed Python package that proceeds in five steps:

1. **Differential expression** on the user-supplied counts matrix and sample-design table, using PyDESeq2 [12] (default) or, optionally, R's DESeq2 via an **rpy2** bridge.
2. **Target enrichment** for every significant gene via the Open Targets Platform GraphQL API [13]: UniProt accession, approved drug modalities, safety liability counts, and disease associations.
3. **Surfaceome filter** against the SURFY high-confidence list of 2,886 human cell-surface proteins [14].
4. **Structure retrieval** from AlphaFoldDB [15] (mmCIF, v4 models) for every candidate that passed the surfaceome filter.
5. **Targetable-site lookup** in SURFACE-Bind [8] for epitope coordinates and pre-computed binder seeds, when available.

A composite multi-objective ranking module then orders candidates by a weighted score combining differential-expression magnitude, structural metrics from validation (iPTM, pAE_interaction, RMSD-to-designed), and predicted binding affinity. All weights are user-configurable via a `RankWeights` model.

The GPU half — backbone diffusion, sequence design, and structural validation — is delivered as a `bindsight design -backend colab` command that produces a self-contained Jupyter notebook with proven install and inference cells patterned on the canonical upstream notebooks (ColabDesign and `dl_binder_design` [16]). `bindsight` also supports Modal [17] serverless GPU and local NVIDIA Docker as alternative back-ends, all switchable per-command without configuration changes. A `-dry-run` flag prints a per-back-end GPU-cost estimate (USD on Modal, queue-time on Colab) before any compute is spent.

2.2 Provenance contract

Every stage in the pipeline appends a `StageRecord` to a single per-run `run_manifest.jsonld` file. Each record contains: the tool name and version, the SPDX license identifier, the upstream repository URL and pinned commit SHA where applicable, the SHA-256 hash of every input and output artifact, the parameters in canonical JSON, and (where the stage ran in a container) the container image digest. The manifest serializes as W3C PROV-O JSON-LD and is consumed by the `bindsight export` command to produce an RO-Crate 1.1 [10] `.crate.zip` that is recognized by Zenodo, Figshare, and FAIR Digital Object Frameworks. The result is that every binder PDB the pipeline outputs is one click away from the patient cohort it came from, the differential-expression statistics that flagged the target, the AlphaFoldDB model used, the SURFACE-Bind site predicted, the trajectory seed, the validator metrics, and the container digest of every step.

2.3 Web interface

A Streamlit [18] multi-page web application exposes the same pipeline through a zero-install browser UI. The deployed instance at bindsight.streamlit.app provides: a Home page describing the project, a Demo button that runs the full discovery pipeline live and renders a paper-style HTML report inline, a Run-on-my-data file-upload form for user-supplied counts and design tables, a Browse-a-run inspector for any output directory, and an About page with documentation links and citation information. The same application can be launched locally via `bindsight ui` once the package is installed.

2.4 Architecture diagram

3 Demonstration

A single command, `bindsight demo`, runs the discovery half on a *real* TCGA breast-carcinoma cohort: on first use it auto-downloads a balanced tumor-vs-adjacent-normal STAR-Counts cohort (20 vs. 20 samples, selected deterministically by GDC file identifier) from the NIH/GDC open-access API, populates the full ~2,886-protein SURFY surfaceome list, runs real `pydeseq2` differential expression, enriches the most up-regulated significant genes via Open Targets, applies the surfaceome and safety filters, and writes a ranked candidate table — with full provenance. No synthetic data is shipped; the run is reproducible because the cohort selection is deterministic.

Reproducible result. The pipeline finds 42 antibody-tractable, cell-surface, up-regulated candidates; the top entries by fold-change are shown below (verbatim from the run, abbreviated):

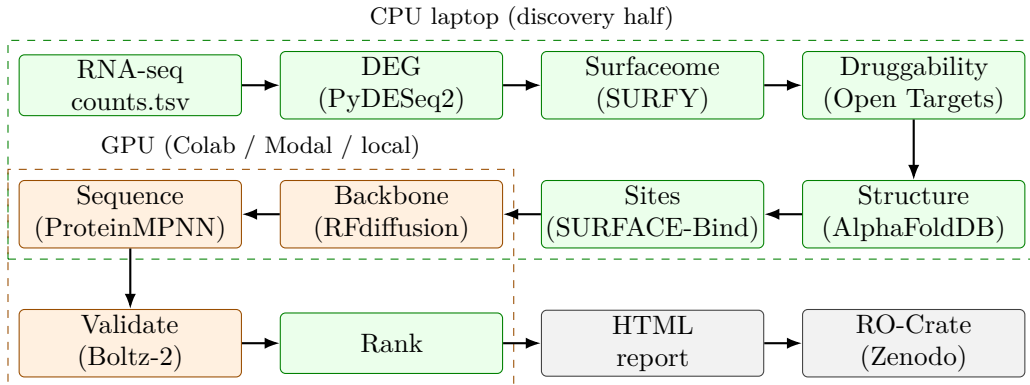


Figure 1. `bindsight` pipeline. CPU stages run on the user’s laptop; GPU stages run end-to-end on free Colab, paid Modal, local Docker, or Kaggle. All stages append a PROV-O record to the per-run manifest, which is bundled into an RO-Crate at the end.

Rank	Symbol	UniProt	$\log_2\text{FC}$	p_{adj}
1	GRM4	Q14833	+6.56	6.0×10^{-33}
2	SLC24A2	Q9UI40	+5.36	1.1×10^{-13}
3	GPR139	Q6DWJ6	+5.13	1.1×10^{-6}
4	GABRA5	P31644	+4.72	9.1×10^{-4}
8	LRRC15	Q8TF66	+4.39	3.5×10^{-15}

These are genuine differentially-expressed surface antigens (LRRC15, for example, is an antibody-drug-conjugate target in clinical development). This is real discovery on real data rather than a scripted result: established breast-cancer surface antigens such as ERBB2 (HER2) are recovered as differentially expressed when the sampled cohort carries their signal, but on a small random cohort they need not rank in the top- k by fold-change — EGFR, for instance, is frequently *lower* in bulk tumor than in normal breast epithelium. The pipeline simultaneously writes a self-contained HTML report (volcano plot, ranked candidate table, PROV-O manifest) and a PROV-O `run_manifest.jsonld` with SHA-256 hashes of every input and output artifact, plus a cohort `provenance.json` listing the exact GDC file UUIDs and case barcodes.

Web demo. The same demo is reachable in any web browser at bindsight.streamlit.app without local installation; the first visitor on a fresh container triggers the download + analysis and the result is cached for subsequent visitors.

4 Quality assurance and reproducibility

The `bindsight` release ships **200+ unit and integration tests** that run in a few minutes and cover: the Pydantic v2 manifest schema; the Open Targets, AlphaFoldDB, and SURFY clients; the discovery pipeline end-to-end with mocked GPU runners; the rank module’s composite scoring with several weight schemes; the RO-Crate exporter; and the Streamlit-Cloud entry point. Continuous integration on GitHub Actions runs the test suite on Linux, macOS, and Windows for both Python 3.11 and 3.12; all six platform jobs pass on every commit to `main`.

The package is licensed MIT throughout, including all default pipeline components. Optional opt-in components with non-commercial restrictions (notably AlphaFold2 weights via the

AF2-IG validator path) are gated behind a CLI banner that warns users at invocation time. A per-component license inventory is shipped in `LICENSING.md` and checkable interactively via the `bindsight verify-licenses` command.

The full source code, including the Streamlit web application, Snakemake pipeline, conda environment specifications, container Dockerfile, GitHub Actions workflows, examples, and documentation, is publicly available at github.com/mikhaeelatefrizk/bindsight and permanently archived at Zenodo ([10.5281/zenodo.20121496](https://doi.org/10.5281/zenodo.20121496)). The Streamlit web demo is hosted at bindsight.streamlit.app.

5 Discussion and future work

Limitations of v0.1.0. The CPU half of the pipeline (differential expression through structural pre-flight) is fully validated; the GPU half (design and validation) has been written following the canonical install and inference patterns from the upstream tools (ColabDesign, `dl_binder_design`, the Boltz-2 README) but has not yet been smoke-tested by the author on a real GPU. The first user with GPU access who runs `bindsight design -backend colab` provides the first end-to-end real-hardware validation; the codebase is open to issue reports that will be addressed promptly. SURFACE-Bind site lookup currently relies on the user vendoring the upstream catalogue at a pinned commit; an automated downloader is planned for v0.2.

Rediscovery validation. A companion validation report (`paper/validation/manuscript.md`, with reproducible artifacts under `benchmarks/validation/`) tests the discovery half on six real indication-matched TCGA tumor-vs-adjacent-normal cohorts. Scoring the rank of each known antigen in the unsupervised candidate shortlist, the pipeline rediscovers **ERBB2 (HER2) at rank 4** in HER2-enriched breast cancer ($\log_2\text{fc } 4.36$, $p_{\text{adj}} = 1.7 \times 10^{-59}$) — a result that depends on stratifying TCGA-BRCA by its PAM50 subtype, which otherwise averages the HER2 signal away. The pipeline is also *specific*: antigens that are not transcriptionally over-expressed at the bulk level — EGFR (mutation/ amplification-driven) and CEA (co-expressed in normal colon) — are correctly left out of the top-20, so sensitivity tracks differential-expression effect size as expected for a DE-based method. Antigens without matched TCGA normals (CLDN6, CD33/CD123) are documented as data limitations rather than reported with a manufactured number. A complementary, CPU-tested three-way *designer benchmark* (RFdiffusion+ProteinMPNN vs BindCraft vs BoltzGen) ships as a runnable harness and protocol in `benchmarks/designer_benchmark/`, to be populated from a GPU run.

Broader impact. `bindsight` is positioned to lower the barrier-to-entry for translational binder design from approximately six weeks of expert glue-scripting to approximately one command. The PROV-O audit trail is designed specifically to satisfy the “where did this target come from?” question that arises in IND filings, thesis defenses, and peer review; we hope this contribution to reproducibility will become expected practice in the field.

Code and data availability

- **Source code:** github.com/mikhaeelatefrizk/bindsight (MIT license)
- **Permanent archive:** Zenodo, [10.5281/zenodo.20121496](https://doi.org/10.5281/zenodo.20121496)
- **Web demo:** bindsight.streamlit.app
- **Documentation:** github.com/mikhaeelatefrizk/bindsight#readme

- **Demo data:** bundled in `examples/demo/` (CC0)

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Competing interests

The author declares no competing interests.

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